

BARKLEY, KY, USA

WMO: 724350

Lat: 37.056N

Lon: 88.774W

Elev: 126

StdP: 99.82

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-8.9	-17.7	0.8	-10.2	-14.9	1.0	-7.3	10.4	5.4	9.1	5.8	3.0	20	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	34.5	24.6	33.3	24.6	32.3	24.2	26.5	32.4	25.8	31.4	25.2	30.5	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.1	29.8	24.2	19.4	29.2	23.6	18.7	28.5	82.9	32.3	80.2	31.5	77.4	30.4	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.9	DB	-15.6	36.6	3.7	1.7	-18.3	37.8	-20.5	38.8	-22.6	39.8	-25.3	41.0	(4)
(5)			WB	-16.2	27.6	3.6	0.7	-18.8	28.1	-20.9	28.5	-22.9	28.9	-25.6	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	1.9	4.3	9.1	14.9	20.0	24.5	26.0	25.4	21.5	15.4	8.8	4.0		
	DBStd	9.79	6.58	6.28	5.85	4.95	4.20	2.98	2.59	2.62	3.85	4.91	5.37	5.84		
	HDD10.0	841	263	179	88	14	0	0	0	0	9	86	202			
	HDD18.3	2219	509	394	289	124	34	1	0	1	14	119	289	445		
	CDD10.0	2558	13	18	61	161	309	436	496	477	347	176	49	15		
	CDD18.3	894	0	1	4	21	84	187	238	219	111	27	2	0		
	CDH23.3	8762	0	1	26	166	722	1862	2454	2192	1073	257	9	0		
	CDH26.7	3534	0	0	1	21	211	765	1084	971	418	63	1	0		
(14) Wind	WSAvg	2.9	3.6	3.6	3.7	3.6	2.9	2.5	2.2	2.0	2.1	2.6	3.0	3.3		
(15) Precipitation	PrecAvg	1235	84	96	115	125	126	106	106	76	97	83	106	113		
	PrecMax	1664	255	339	379	369	237	279	317	184	234	187	267	293		
	PrecMin	923	15	24	24	39	36	10	7	11	3	0	23	16		
	PrecStd	196	54	60	65	70	58	65	70	42	56	45	62	70		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	21.9	26.2	28.7	32.1	34.8	36.4	36.7	34.5	30.8	24.6	20.8		
		MCWB	16.3	16.8	17.7	19.4	22.8	24.2	25.7	24.3	22.5	21.8	17.7	17.2		
	2%	DB	17.8	19.4	23.5	27.0	30.5	33.3	34.4	34.5	32.8	28.4	22.1	18.0		
		MCWB	15.0	14.9	16.4	18.8	21.9	24.2	25.6	24.5	23.2	20.0	16.2	14.9		
	5%	DB	15.3	17.0	21.3	25.3	29.1	32.3	33.1	33.0	31.1	26.3	20.0	15.7		
		MCWB	12.8	12.9	15.0	17.8	21.2	23.9	25.4	24.5	22.3	18.7	15.2	12.8		
	10%	DB	12.4	14.5	18.9	23.5	27.6	31.0	31.9	31.6	29.1	24.0	17.8	13.3		
		MCWB	9.5	10.8	13.6	16.9	20.7	23.4	24.7	24.0	21.1	17.6	13.6	10.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	17.7	19.0	21.6	24.5	26.6	27.4	27.3	25.2	23.2	19.4	17.6		
		MCDB	18.9	20.9	24.1	26.4	29.8	32.3	33.2	33.0	31.3	29.1	22.6	20.2		
	2%	WB	15.5	15.7	17.5	20.3	23.3	25.7	26.7	26.3	24.1	21.7	17.8	15.6		
		MCDB	17.4	18.4	22.2	25.1	28.4	31.5	32.6	32.1	30.0	26.2	21.0	17.7		
	5%	WB	13.3	13.9	16.0	19.1	22.4	25.0	26.0	25.6	23.4	20.3	16.1	13.4		
		MCDB	15.0	16.4	20.2	23.6	27.5	30.5	31.7	31.0	28.9	24.3	19.3	15.2		
	10%	WB	10.0	11.1	14.4	17.9	21.5	24.2	25.4	24.9	22.5	18.9	14.1	10.8		
		MCDB	12.0	14.0	18.3	22.2	26.2	29.2	30.7	29.8	27.3	22.8	16.9	12.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	10.9	11.7	11.0	10.8	10.5	11.4	12.7	12.4	10.8	8.9		
		MCDBR	10.7	12.6	13.2	12.9	12.1	12.0	11.5	12.9	14.1	14.3	12.7	10.9		
	5% WB	MCWBR	8.9	9.4	8.1	6.8	5.5	4.7	4.5	5.0	5.7	7.0	8.4	8.9		
		MCWBR	9.7	11.1	11.7	10.8	10.4	10.4	10.4	10.8	11.6	11.5	10.8	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.328	0.365	0.400	0.428	0.440	0.478	0.476	0.410	0.360	0.324	0.314		
		taud	2.460	2.413	2.370	2.274	2.251	2.228	2.147	2.140	2.323	2.444	2.490	2.477		
	Ebn at Noon		870	904	899	881	859	845	809	801	840	857	857	850		
		Edh at Noon	84	99	113	132	137	140	151	149	118	95	81	77		
(44) All-Sky Solar Radiation	RadAvg		2.16	2.76	3.84	5.04	5.68	6.38	6.16	5.65	4.74	3.59	2.46	1.83		
	RadStd		0.20	0.34	0.33	0.32	0.48	0.41	0.38	0.29	0.29	0.39	0.26	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+96		
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page